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## ECE3700J Introduction to Computer Organization

### Lab 4 – Resolving Hazards

#### Purpose

This lab is intended to take care of data hazards and control hazards that may arise in pipelined processors by adding forwarding mechanism, hazard detection components, and signals for stalls in hardware.

#### Tasks

This lab is based on the pipelined processor lab. You will add the necessary forwarding paths, hazard detection components, and stall mechanisms to the pipelined processor designed in previous lab to avoid stalls in software program. For data hazard, consider the cases that have been discussed in the lectures only. For control hazard, assume branch is always not taken. The entire design should be modeled in Verilog HDL.

The design must be simulated with a Verilog simulator of your choice and synthesized by using Xilinx synthesis tool, meeting the following requirements:

- Simulation results must be error-free
- An RTL schematic must be produced from the synthesis tool

FPGA hardware implementation is NOT required for this lab.

A simple RISC-V assembly testing program with data and control hazards will be provided for you to verify that your processor can execute those instructions continuously and correctly.

#### Team Organization

This lab is a team effort. Each team should consist of 3 students, randomly grouped. The work should be appropriately divided and distributed among all team members. Every student will be given oral examination upon demonstration. Students are not allowed to switch teams without permission of the instructor.

#### Deliverables

- **Demonstration** – Every team should demonstrate to the teaching group the following before your lab session ends:
  - 1) Simulation results of your design by executing the given RISC-V assembly program;
  - 2) RTL schematic generated by the synthesis tool.Each team member should be prepared for an oral exam on this lab during the demonstration.
- **Lab report** – The lab report should be a written report including:
  - 1) Brief description of all aspects of the modeling and implementation of the processor;
  - 2) Screen shots and brief explanations of simulation results for all potential cases of data hazard and control hazard.
  - 3) RTL schematic of your Verilog model generated with Xilinx Vivado software;



- **Peer Evaluation** – Each team member is required to provide a peer evaluation for the team effort in this lab. The marks of the peer evaluation should be integers ranging between 0 to 10, inclusively, with 10 indicating the biggest contribution. A mark should be given to each team member including yourself according to the team member's contribution based on your observation. A brief description of contribution of each team member should also be provided, as shown in the following table.

| Name                  | Level of contribution<br>(0 ~ 10) | Description of contribution |
|-----------------------|-----------------------------------|-----------------------------|
| (yourself)            |                                   |                             |
| (your lab partner 1 ) |                                   |                             |
| (your lab partner 2)  |                                   |                             |

- **Source Files** – All your Verilog source files and any other supporting files.

This is a 1-week lab. The full score for this lab is 500 points.

All required documents should be submitted on Canvas before **22:00pm, July 12, 2025**.

## Grading

- Lab report: 30%
- Demonstration: 50%
  - Working Verilog model (simulation): 30%
  - Individual oral exam: 20%
- Source files and peer evaluation: 20%